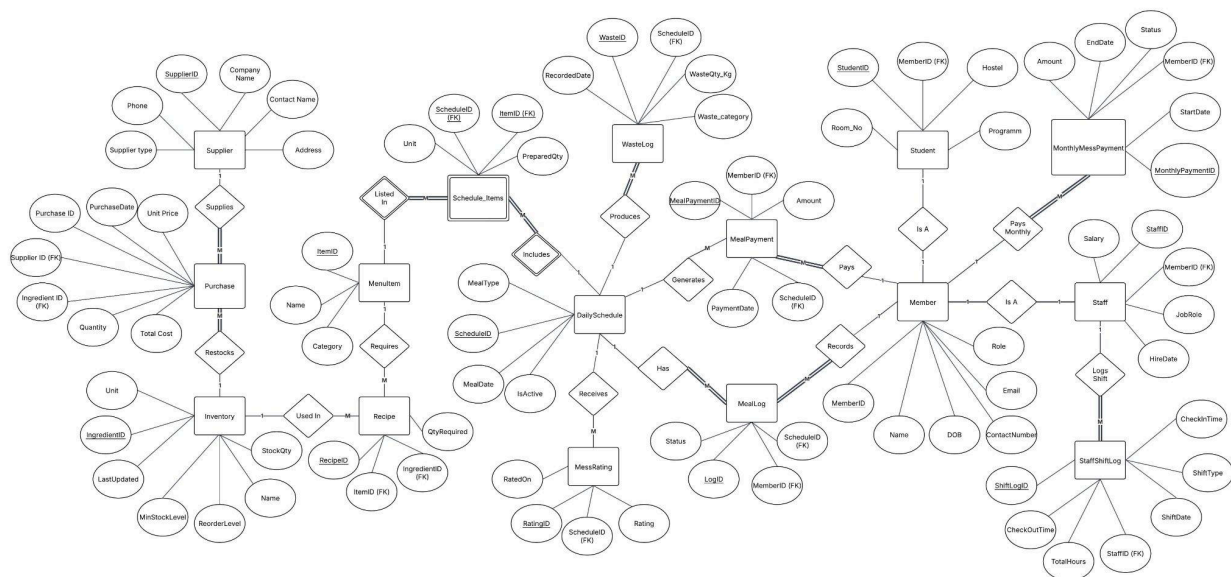




2. Explanations & Justifications

UML to ER Mapping Transition

The transition from the Object-Oriented design to the Relational ER model follows these rules:

1. **Inheritance Mapping:** The Member table contains common attributes. Student and Staff tables link to it via MemberID (1:1 relationship) to handle specialization.
2. **M:N Relationships:** The relationship between DailySchedule and MenuItem is resolved by the Schedule_Items table, which tracks `QuantityPrepared` for each item in a meal.
3. **Aggregation:** Recipe links MenuItem and Inventory, defining ingredients required for dishes.

Relationship Justifications

- **Payment Separation:** We separated MonthlyMessPayment (for regular fees) and MealPayment (for individual meal purchases) to handle different billing cycles effectively.
- **Staff Management:** StaffShiftLog tracks working hours precisely with check-in/out times, linked to the main Staff entity.

Constraints & Integrity

- **Primary Keys (PK):** Uniquely identify each record. Marked clearly in the diagram.
- **Foreign Keys (FK):** Establish relationships between tables. All Foreign Keys use `ON DELETE CASCADE` (as seen in the SQL) to ensure data consistency.
- **Domain Checks:** Constraints like Salary > 0 and Status enums are enforced via SQL CHECK constraints, while logical domain constraints like an Age restriction (Age >= 16) are derived dynamically from the DOB attribute.

3. Team Members & Contributions

Team Name: Interview Rejected

Name	Contribution
Sai Teja [23110366]	ER Diagram, Wrote SQL Queries.
Moksh Jindal [23110210]	Wrote SQL Queries, Designing of Database.
Mohit Yadav [23110209]	Designing of Database.
Tejas Patil [23110333]	Compiled the report.
Ankit Singh	NA